Session 4: Reproducible Documents

Elisabeth Thamm

Table of contents

esc-r makes a raw cell, just plain text (markdown and code are executed)

1 is a heading, $ equation $$ equation on seperate line

2 Rocket Equation

At time t_0 to rocket starts to expel gas at a *constant mass flow rate* R meassured in kg/s and *exhaust velocity relative to the rocket* v_e in m/s.

$$\frac{dv}{dt} = -\frac{F}{m(t)} = -\frac{Rv_e}{m(t)} \tag{1}$$

Integrating both sides of Equation 2 from 0 to T we get

3 Rocket Equation

At time t_0 to rocket starts to expel gas at a *constant mass flow rate* R meassured in kg/s and *exhaust velocity relative to the rocket* v_e in m/s.

$$\frac{dv}{dt} = -\frac{F}{m(t)} = -\frac{Rv_e}{m(t)} \tag{2}$$

Integrating both sides of Equation 2 from 0 to T we get

Open data in python: (Reading a Data File)

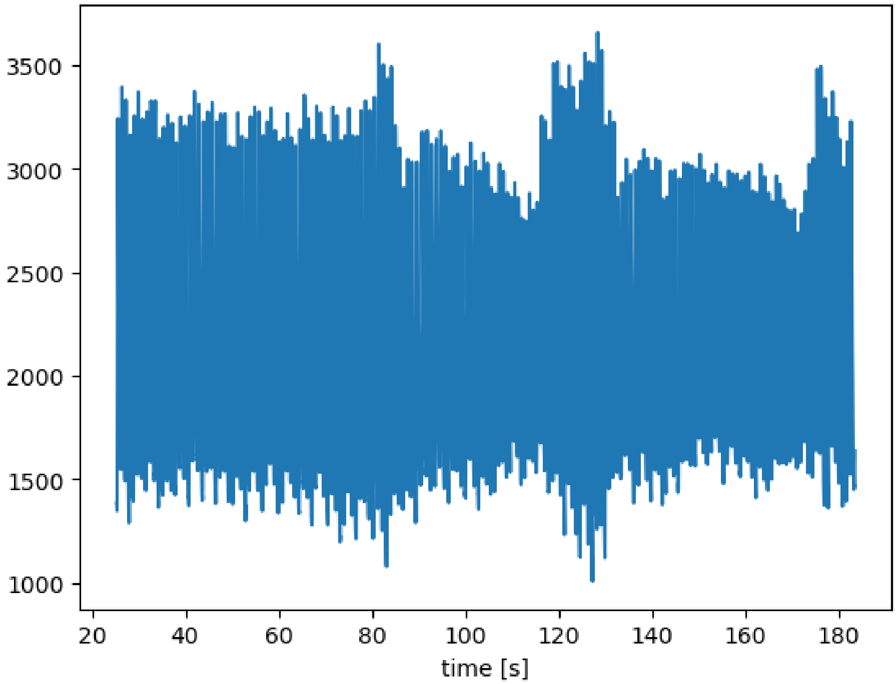
create an empty list:

Numpy is on the Charité server, for calculating things like matlab; matplotlib is another library, pyplot is a collection within

Homework session 3 removed

```
Text(0.5, 0, 'time [s]')
```

(a) Measurements from an oxymetry sensor as a function of time for 3 cycles



(b)

Figure 1